

CCN_Emotion — Project Flow

Brain-predictable subspace of a self-supervised video model

① INPUT

Horikawa 2020 fMRI
5 subjects × 2,196 emotional videos
(Cowen-Keltner 34 categories)

Brain-JEPA (NeurIPS 2024)
fMRI foundation model
768-dim subject-invariant

V-JEPA2 (Assran 2025)
Self-supervised video model
ViT-G, 1408-dim, no emotion labels

② CORE METHOD

V-JEPA2 → PCA (100 components)
variance-based dimensionality reduction

Brain-JEPA → V-JEPA2 PC ridge regression
5-fold CV + permutation test (n=1000)
FDR correction ($q < 0.05$)

③ FINDING

3 PCs survive FDR
PC1 $R^2=0.373$ (dominant)
PC2 $R^2=0.075$ (marginal)
PC3 $R^2=0.088$ (marginal)

Inside brain-predictable subspace
category mean $R^2 = 0.055$
V-A mean $R^2 = 0.038$
ratio = 1.44 (vs full 100-PC = 1.26)

④ TWO INTERPRETIVE LEAPS (paper claim ↔ honest evidence)

HONEST EVIDENCE
3 visual feature axes brain tracks;
videos cluster more by category label
than by arousal-valence dimensions

LEAP 1
visual axes → "affective"

PAPER CLAIM (over-reach)
"affective subspace ... SSL spontaneously
produces categorically organized
affective representation"

LEAP 2: cite Kragel (2019) + Conwell (2025)
as mechanism — but Kragel = supervised,
Conwell = behavior only. Affirming consequent.

⑤ THREE PILLARS TO VALIDATE

PILLAR 1 — EXISTENCE ✓
Low-dim brain-readable subspace
(3 PCs survive FDR)
ALREADY SHOWN

PILLAR 2 — SPECIFICITY (to do)
Partial-out DINOv2 (object),
Places365 (scene), optical flow (motion),
Sadeghi 139 low-level. Does ratio survive?

PILLAR 3 — SSL CONTRIBUTION (to do)
Untrained ViT, ImageNet-supervised ViT-L,
VideoMAE. Does only V-JEPA2 show the
ratio, or do all visual models?

⑥ THREE OUTCOME SCENARIOS

(a) TRIVIAL
Fully explained by
standard visual recognition.
CCN poster only. Camera-ready softening.

(b) ANY VISION MODEL
Residual structure exists,
but not SSL-specific.
Conwell 2025 brain version.

(c) SSL EMERGENT
Residual structure exists
AND only in SSL video pretraining.
NeurIPS / Nature Comm candidate.

BENCHMARKS:

Horikawa et al. (2020, iScience) — same dataset, same question, primitive 2020-era visual baseline.
Sartzetaki et al. (2025, ICLR) — across-model alignment decomposition. We are the within-model dual.